SHAPES *with*

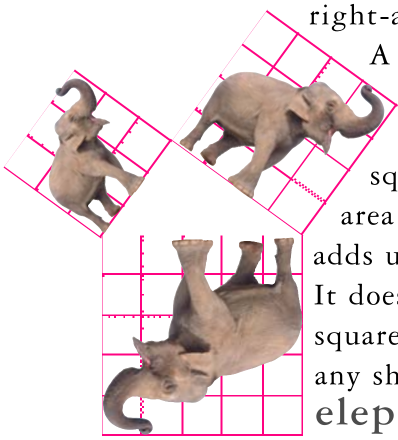
Shapes made of straight lines are called **polygons**. The simplest polygons are triangles, which are made from three straight lines joined at three corners, or angles. Triangles are the building blocks for all other polygons.

Triangles can cover a flat surface *completely* without leaving gaps

THE RIGHT STUFF

Mathematicians' favourite triangles are those with one L-shaped corner: **right-angled triangles**.

Ancient Egyptians used right-angled triangles to make square corners to mark out fields or buildings. They knew a loop of rope with 12 equally spaced knots made a right-angle if you STRETCHED it into a triangle with sides 3, 4, and 5 knots long.



The ancient Greeks knew about right-angled triangles too. A man called **Pythagoras** discovered something special about them: if you draw squares on each side, the area of the two small squares adds up to the big square. It doesn't just work for squares, it works for any shape, even **elephants**!

So what? Pythagoras's discovery became the most famous maths rule ***of all time***. Pythagoras was apparently delighted with it – according to legend, he celebrated by sacrificing an ox.

No matter what shape a triangle is, the three angles always add up to 180°. Here's an ingenious way of proving it:

1. Use a ruler to draw a large triangle on a piece of paper. Then cut it out.

2. Tear off the three corners...

3. ...and put them together like this →

They'll always form a straight line, which proves the angles add up to 180°.